
INTERNSHIP PROJECT SUBMISSION

Heart Disease Prediction

Predicting risk. Supporting earlier action.

SEPTEMBER 12, 2026



A model built for earlier intervention.

The project turns routinely collected clinical attributes into a practical heart-disease risk prediction workflow.

PURPOSE

Identify heart disease risk with a predictive model.

IMPORTANCE

Earlier detection can save lives and reduce costs.

SCOPE

Analysis, model building, evaluation and validation.

THE SCALE OF THE CHALLENGE

31%

of global deaths

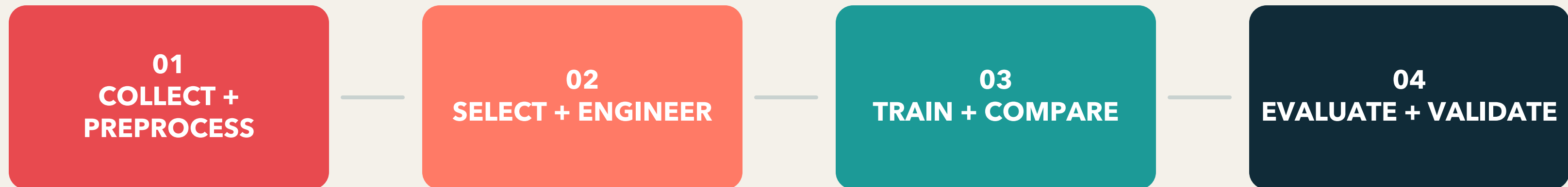
Heart disease remains the leading cause of death worldwide – making timely risk identification an urgent public-health priority.

GLOBAL HEALTH CONTEXT



From raw records to validated predictions.

A four-stage, reproducible machine-learning workflow



03 / IMPLEMENTATION

Built in a familiar, reproducible stack.

PYTHON • JUPYTER • SCIKIT-LEARN • PANDAS

DATASET

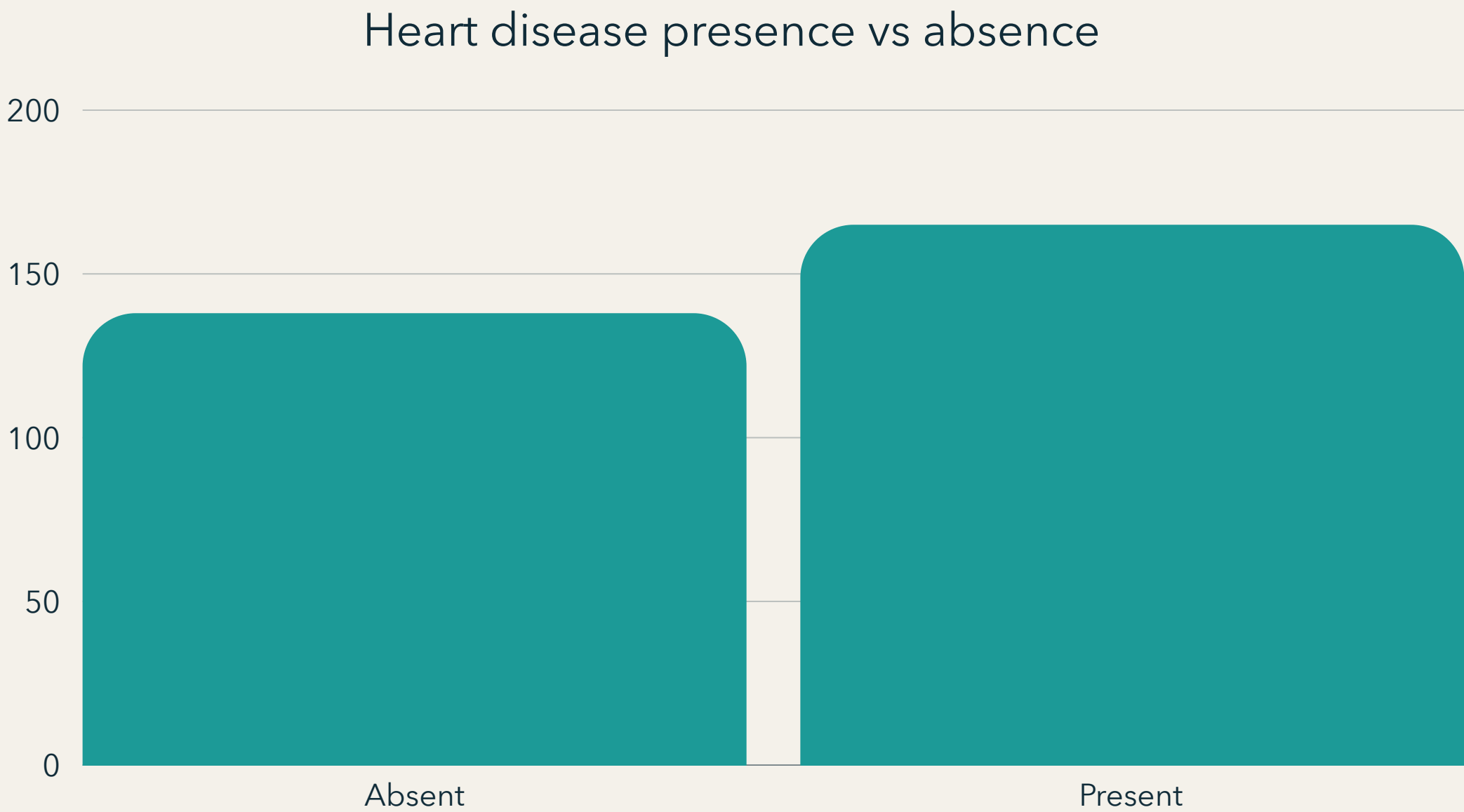
UCI Heart Disease Dataset

303 patient records / 14 clinical attributes



303 patients. 14 attributes.

Age, sex, blood pressure, cholesterol, maximum heart rate and other clinical indicators.



303
PATIENT RECORDS

14
CLINICAL FEATURES

Three contenders. One clear choice.

LOGISTIC REGRESSION

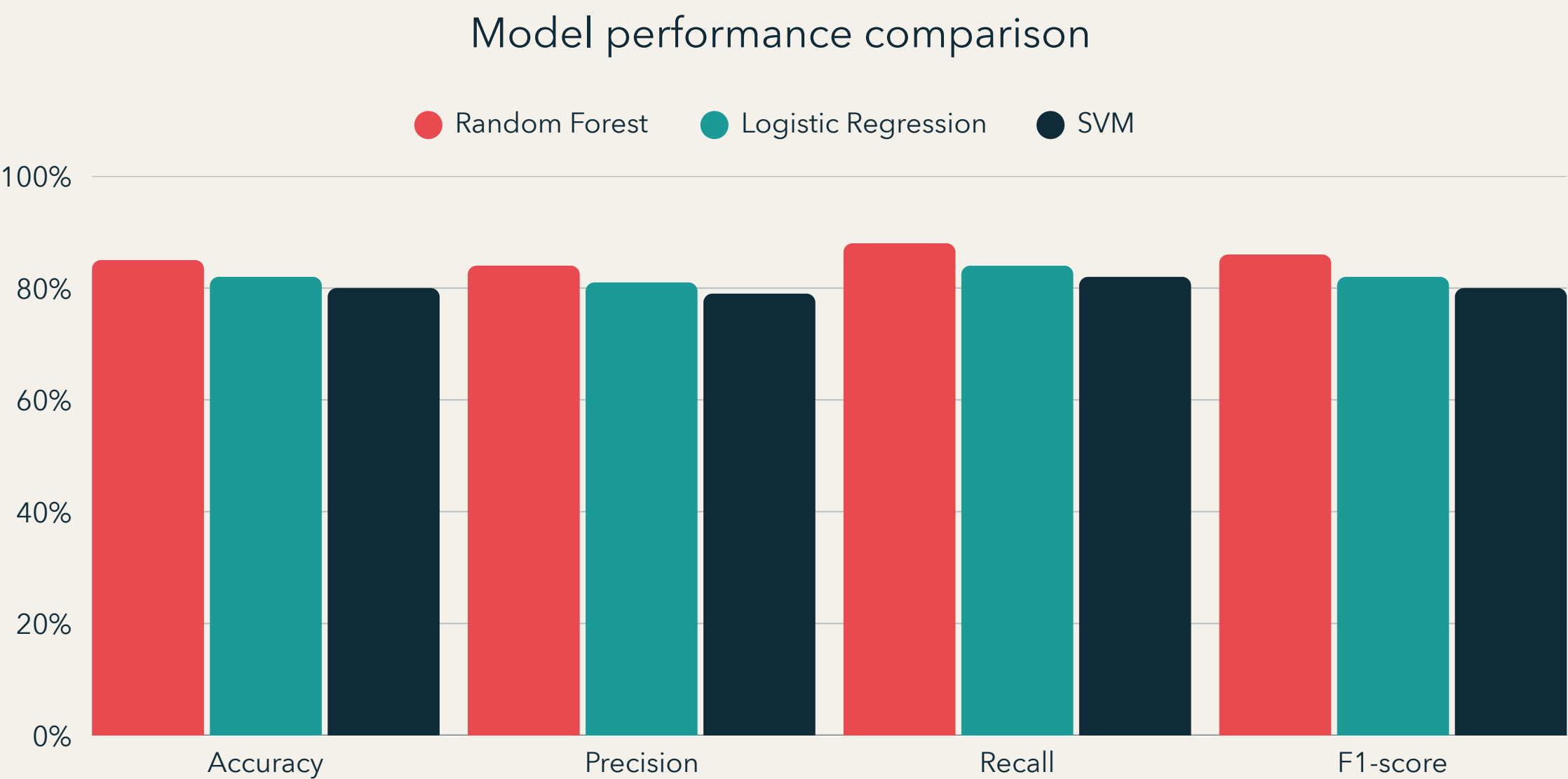
SUPPORT VECTOR MACHINE

RANDOM FOREST

85%

FINAL MODEL ACCURACY

Random Forest leads across the measures that matter.



CONFUSION MATRIX / RANDOM FOREST

68 TRUE NEG.	7 FALSE POS.
9 FALSE NEG.	67 TRUE POS.

What the model revealed.

01

Random Forest won

It delivered the strongest accuracy and recall.

02

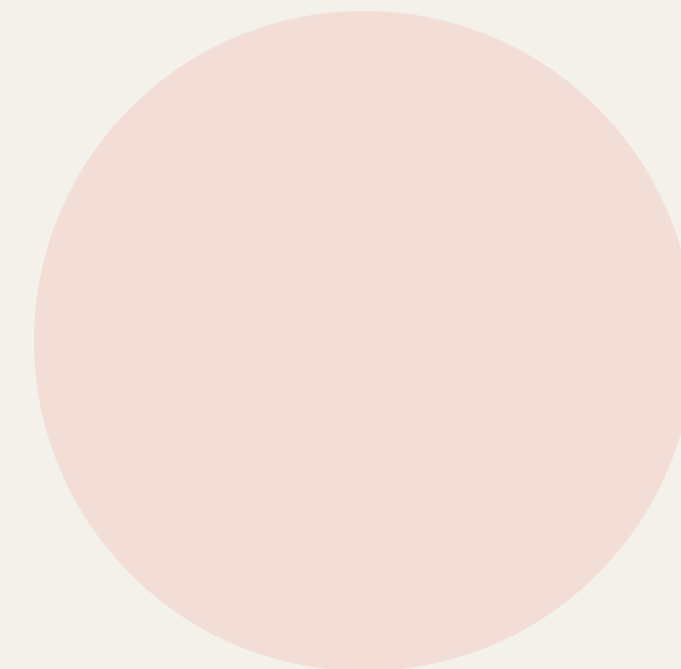
Signals stood out

Age, cholesterol and maximum heart rate ranked highest.

03

Preparation mattered

Preprocessing materially improved model robustness.



The work behind the result.

Reliable prediction depends as much on disciplined validation as it does on model choice.

CLEAN

Handle missing data and scale features consistently.

BALANCE

Trade model complexity against interpretability.

VALIDATE

Use cross-validation for results that travel beyond one split.

HEART DISEASE PREDICTION

Thank you.

Questions and feedback welcome.

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